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Data Assignment 2 Report

1. Introduction

Selective serotonin reuptake inhibitors (SSRIs) are commonly prescribed for treating depression, but the timing, likelihood, and key factors influencing remission in real-world patients remain unclear. The STAR*D study, which is what is being used in this assignment, aimed to assess the effectiveness of citalopram, an SSRI, in routine clinical practice using measurement-based care and to determine factors associated with symptom remission in outpatients diagnosed with major depressive disorder.

2. Methods

2.1 Data Preparation

The STAR*D study focused on individuals diagnosed with major depressive disorder who were receiving outpatient treatment. These patients were recruited from a mix of psychiatric clinics and primary care offices, reflecting typical, non-research clinical environments. The study included patients who were prescribed citalopram as part of their routine care.

2.2 Data Exploration

Spaghetti plots were used to visualize individual subject changes over time and to assess the potential for subjects to have similar baseline depression levels. Given the known variability in depression severity, it was anticipated that subjects would likely start at different levels. To examine the overall trend, loess curves were applied, which can accommodate potential non-linearity in the data. As gender and race may influence both baseline depression levels and recovery rates, plots were generated to compare these groups.

2.3 Statistical Modeling

To investigate the factors influencing QIDS scores over time, three linear mixed-effects models were considered. The **baseline model** included 'WEEK' as the sole fixed effect predictor, with random intercepts and slopes for 'WEEK' to account for individual variability in baseline QIDS scores and rates of change. This model served as a fundamental representation of the temporal change in QIDS. The **extended model** expanded upon the baseline by incorporating additional fixed effects: gender, race, scaled age at diagnosis ('dage_scaled'), scaled episode number ('epino_scaled'), and family psychiatric history ('Any_Fam_Psyc_Hist'). This model aimed to assess the impact of demographic and clinical variables on QIDS scores. Finally, the **custom model** was designed to provide a more comprehensive and theoretically grounded analysis. This model included 'WEEK' as a fixed effect, along with a composite variable 'Total_Fam_Psyc_His' (summing indicators of depression, substance use disorders, and bipolar disorder) to capture family psychiatric history. Scaled age at diagnosis ('dage_scaled'), scaled episode number ('epino_scaled'), race ('Race_Collapsed'), and scaled episode date ('episode_date_scaled') were also included. Random intercepts and slopes for 'WEEK' were specified for each participant to account for individual variability. This custom model was chosen to balance complexity and interpretability, capturing both individual and group-level influences on QIDS scores while avoiding excessive multicollinearity. The 'Total_Fam_Psyc_His' variable was engineered to summarize key aspects of family psychiatric history.

Model selection was based on a comparison of Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and log-likelihood values. Lower AIC and BIC values, along with higher log-likelihoods, were used to identify the model that best balanced goodness of fit and parsimony. This approach aimed to select the model that most accurately represented the data while avoiding unnecessary complexity.

3. Results

Table 2

Baseline Characteristics of Patients		
Characteristic	**%**	**N**
Race		
Asian	1.7	56.0
Black	16.9	559.0
Other	1.4	47.0
White	80.0	2,642.0
Gender		
Female	61.6	2,034.0
Male	38.4	1,270.0
Depression History		
No	44.6	1,475.0
Yes	54.6	1,803.0
Alcohol Use		
No	59.3	1,960.0
Yes	39.9	1,319.0
Drug Use		
No	76.1	2,513.0
Yes	23.2	766.0
Suicidal History		
No	95.9	3,167.0
Yes	3.2	107.0
Characteristic	**IQR**	**Median**
Age	21.0	21.0
Major Depressive Episodes	4.0	3.0
Length of Current MDE	641.0	247.0

3.1 Descriptive Statistics

Two demographic datasets, STARD Demo 1.csv and STARD Demo 2.csv, were merged using the ID variable to consolidate patient information. Only subjects with at least three recorded visits were retained for analysis. For simplicity, any visits that have .1 in them were deleted. The frequency of visits by patients were counted as shown in Table 1. Negative values in the dataset were examined and handled appropriately to ensure data integrity. In this analysis, subjects were allowed to select multiple races, and it was necessary to classify them into broader categories for reporting purposes. The following classification rules were applied to categorize race: Subjects who identified as White were classified as White. Subjects who identified as Black and did not identify as White were classified as Black. This ensures that individuals who selected both White and Black are not mistakenly categorized as Black. Subjects who identified as Asian and did not identify as either Black or White were classified as Asian. Subjects who selected races such as Hispanic, AMIND, or HAWAI, and did not select White, Black, or Asian, were classified as Other. This classification approach ensures that individuals who selected multiple races are categorized based on their most specific race (e.g., Black if Black and White were selected). Additionally, the Other category includes subjects who selected races outside the main classifications, ensuring that their race is still represented in the dataset without misclassification. Patient characteristics can be observed in Table 2.

Visits	3	4	5.5	7	8
Frequency	298	401	621	1161	824

Table 1

3.2 Data Visualizations

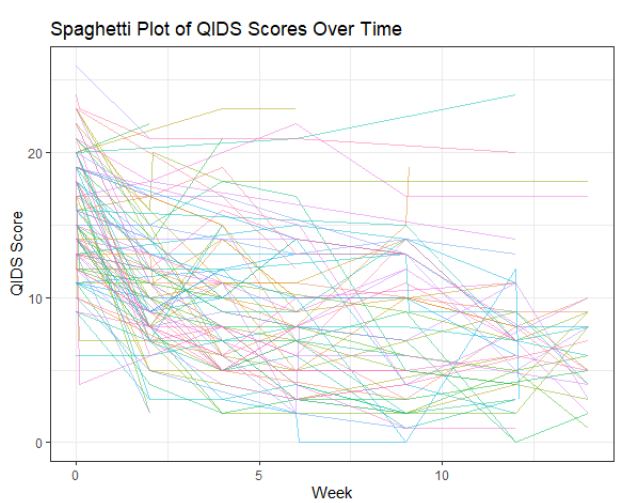

Figure 1

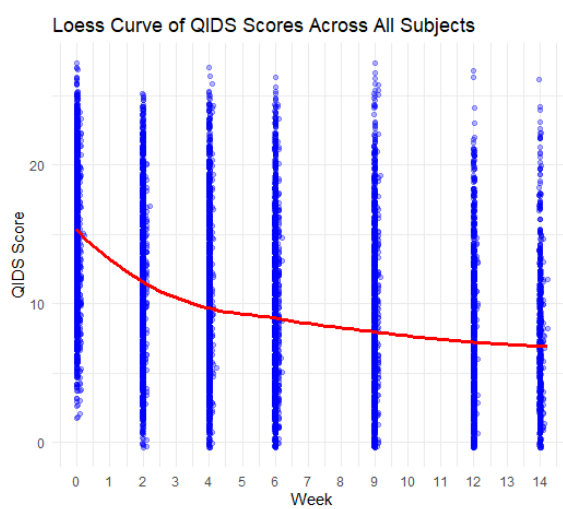

Figure 2

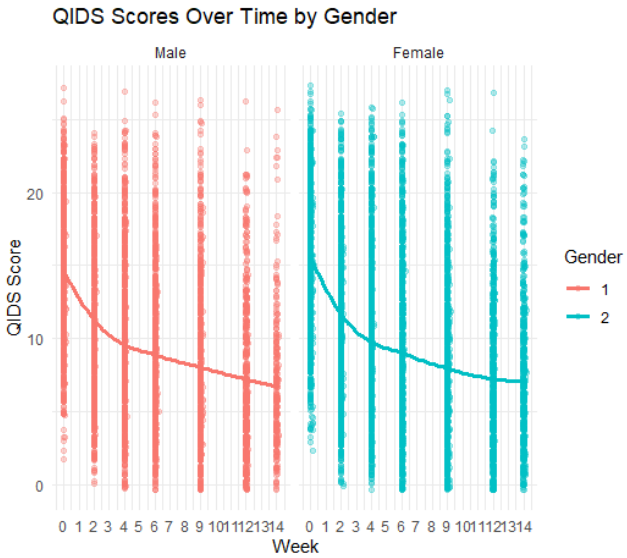

Figure 3

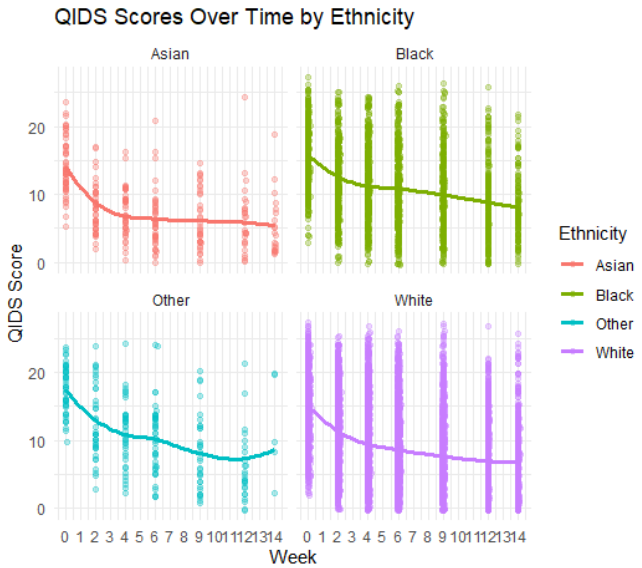

Figure 4

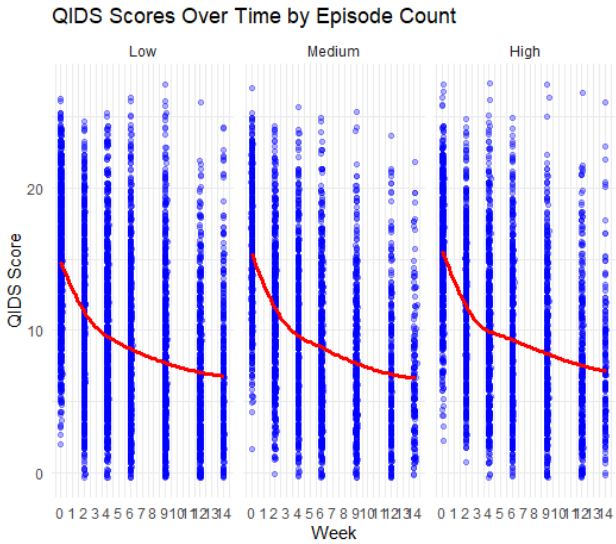

Figure 5

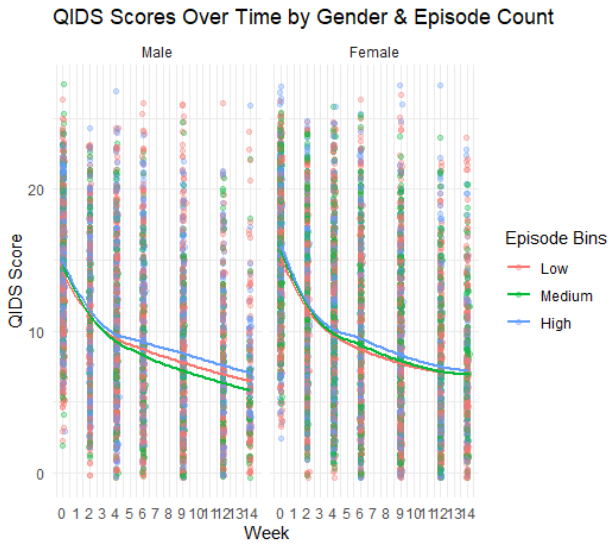

Figure 6

The effectiveness of citalopram treatment for major depressive disorder can be visualized across several dimensions in these figures. The Spaghetti Plot of QIDS Scores Over Time (Figure 1) illustrates the individual variability in patient response, with a general trend toward lower depression scores over the study period, although some patients show minimal improvement or fluctuating symptoms. This overall trend is reinforced in the L[illegible]Curve of QIDS Scores Across All Subjects (Figure 2), which depicts the average decrease in QIDS scores, offering a clearer view of the treatment's average effect. When examining potential moderating factors, QIDS Scores Over Time by Gender (Figure 3) and QIDS Scores Over Time by Ethnicity (Figure 4) allow for comparisons of treatment response across these groups, potentially revealing differences in baseline severity or rates of improvement. The distribution of change scores provides further insight into treatment outcomes; Distribution of HAMD Change Scores (Figure 5) shows the range and frequency of changes in the primary outcome measure, while Distribution of QIDS-SR Change Scores (Figure 6) does the same for the self-report measure, enabling a comparison of how consistently patients' symptoms changed across different assessment tools. Analyzing these distributions in Figures 5 and 6, in conjunction with the definitions of remission and response, would allow for an estimation of the proportion of patients achieving these outcomes.

3.3 Model Comparison

Model 1 (Baseline Model): The fixed effects analysis revealed that the average baseline QIDS score at Week 0 was 13.54. Furthermore, the model indicated a significant decrease in QIDS scores over time, with an average reduction of 0.59 points per week, as evidenced by a large t-value of -70.57, suggesting a strong effect of the 'WEEK' variable. The random effects component of the model highlighted substantial variability among subjects. Specifically, the intercept variance of 12.31, with a standard deviation of 3.51, suggests that individuals began the study with a wide range of initial QIDS scores. The slope variance for 'WEEK', at 0.078 (SD = 0.28), indicates some, albeit relatively small, variation in the rate of improvement across subjects. The residual variance of 10.85 (SD = 3.29) represents within-subject fluctuations in QIDS scores. Notably, the intercept-slope correlation of 0.02 suggests virtually no relationship between a patient's initial QIDS score and their subsequent rate of improvement. Finally, the model fit was assessed using AIC and BIC, with values of 93562.5 and 93608.8, respectively, which will serve as a baseline for comparison with an expanded model. View Appendix B for model output.

Model 2 (Extended Model): The statistical model reveals several key findings regarding factors influencing QIDS scores, a measure of depression severity. At baseline (Week 0), the average QIDS score was 10.65. Over time, there was a significant decrease in depression symptoms, with QIDS scores decreasing by an average of 0.60 points per week. Gender also played a role, with males exhibiting QIDS scores 0.53 points higher than females, on average. Race showed a notable impact, with Black subjects having 3.37 points higher QIDS scores, individuals from other racial backgrounds having 3.89 points higher scores, and White subjects having 1.81 points higher scores, all compared to the reference racial group. Age had a minor influence, with QIDS scores decreasing by 0.01 points for each one-year increase in age. Epilepsy history and family psychiatric history showed positive associations with QIDS scores, but these were not statistically significant at conventional thresholds.

The random effects analysis indicated substantial variability in baseline QIDS scores across subjects (variance = 11.81, SD = 3.44). The rate of change in QIDS scores also varied moderately between individuals (variance = 0.08, SD = 0.28). Additionally, there was considerable unexplained variability within subjects' QIDS scores (variance = 10.85, SD = 3.29). The correlation between gender and race was somewhat negative, suggesting a weak relationship. The correlation between age and other predictors was low, indicating minimal collinearity.

In summary, the model demonstrates a strong negative effect of time on QIDS scores, highlighting a decrease in depression severity over the study period. Race significantly influenced QIDS scores, with Black and Other racial groups showing higher scores compared to the reference group. There was no meaningful correlation between baseline QIDS scores and the rate of change over time, and family psychiatric history did not significantly affect QIDS scores. View Appendix B for model output.

Model 3 (Custom Model): The model, with an AIC of 80240.7 and BIC of 80338.9, reveals that time ('WEEK') has a significant negative effect on QIDS scores, with an average decrease of 0.60 points per week (t = -66.326). This indicates a consistent improvement over the 14-week period. Race plays a substantial role, with Black subjects exhibiting 3.20 points higher QIDS scores (t = 5.589), individuals from other racial backgrounds 3.66 points higher (t = 4.469), and White subjects 1.64 points higher (t = 2.991), all compared to the reference group. Total family psychiatric history is positively associated with QIDS scores, with a coefficient of 0.22 (t = 3.210). The baseline QIDS score is estimated at 11.33 (t = 20.882). Notably, there's significant variability in baseline QIDS scores across subjects, with a variance of 11.81 (SD = 3.44), and moderate variability in the rates of change over time (variance = 0.08, SD = 0.28). However, there's no correlation between baseline scores and the rate of change (correlation = 0.00). While age and the number of major depressive episodes show some association with QIDS scores, these effects are not statistically significant. The correlation matrix reveals high correlations between the different race categories, suggesting these categories are related. Finally, considerable unexplained variability remains within subjects' scores, with a residual variance of 10.76 (SD = 3.28), highlighting the complexity of the factors influencing QIDS scores.

<u>Final Model Selection:</u> Based on the model summaries, Model 3 ("Custom Model") appears to be the most plausible. This conclusion is primarily supported by its AIC and BIC values, which are identical to those of Model 2 but significantly lower than those of Model 1. Lower AIC and BIC values generally indicate a better balance between model fit and complexity. Model 3, like Model 2, incorporates additional predictors beyond just time, such as race and family psychiatric history, but it specifically highlights the inclusion of Total_Fam_Psyc_His as a composite variable. This inclusion seems to improve the model's fit without substantially increasing the residual variance, which remains relatively consistent across all three models. Model 1, the baseline model, while demonstrating a strong effect of time on QIDS scores, has a considerably higher AIC and BIC, suggesting a poorer fit compared to the other two. Therefore, Model 3 is the most suitable, given its improved fit as indicated by the lower AIC and BIC, is well-supported. The model and fitted values for the model can be found in Appendix B.

3.4 Final Model Interpretation

	Estimate <dbl>	Std. Error <dbl>	t value <dbl>
(Intercept)	11.32830484	0.542493614	20.881914
WEEK	-0.59773921	0.009012173	-66.325763
Total_Fam_Psyc_His	0.21998903	0.068534800	3.209888
dage_scaled	-0.13258361	0.076401565	-1.735352
epino_scaled	0.07935822	0.075784616	1.047155
Race_CollapsedBlack	3.20443847	0.573383986	5.588643
Race_CollapsedOther	3.65789837	0.818421993	4.469453
Race_CollapsedWhite	1.63915515	0.548048768	2.990893
episode_date_scaled	0.20747546	0.073081524	2.838959

Figure 7

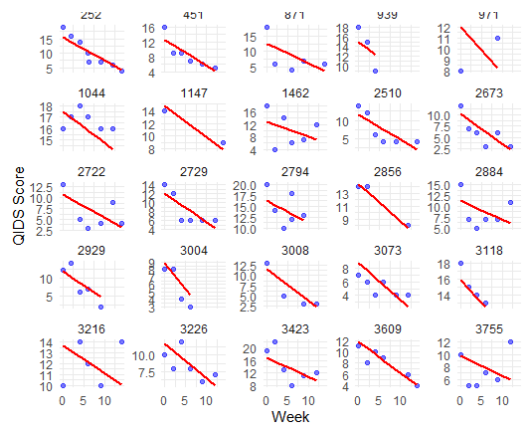

Figure 8

The coefficients for our final model predictors are shown in figure 7. In the model output, the intercept-slope correlation was reported as 0.00. This indicates that there's **essentially no correlation** between a patient's initial QIDS score (intercept) and their subsequent rate of improvement (slope). This implies that a patient's starting point does not predict how quickly they will improve. **Overall Fit:** The red lines generally follow the trend of the blue points, indicating that the model captures the overall pattern of change in QIDS scores over time for most subjects. **Individual Variability:** The plot highlights the substantial individual variability in both baseline scores and rates of change. Some subjects start with high QIDS scores and show a rapid decline, while others start with lower scores and show a more gradual change. **Model Accuracy:** The accuracy of the model's predictions varies across subjects. For some, the red line closely aligns with the blue points, indicating a good fit. For others, there are larger discrepancies, suggesting that the model doesn't fully capture their individual trajectories. **Outliers:** Some subjects may have data points that deviate significantly from the predicted lines, potentially indicating outliers or unique patterns of response. **Slope Variability:** Some subjects have a very steep slope, and some have a very flat slope. Indicating the rate of change is different for each subject. The plot reinforces the importance of using a mixed-effects model, as it accounts for the individual variability in baseline scores and rates of change. The discrepancies between observed and predicted values suggest that other factors not included in the model may be influencing individual patient trajectories. The plot can be used to identify subjects with unusual response patterns, which may warrant further investigation.

4. Conclusion

This analysis examined the factors influencing QIDS scores, a measure of depressive symptom severity, over a 14-week treatment period using a linear mixed-effects model. Key findings revealed a significant decrease in depressive symptoms over time, with an average reduction of approximately 0.60 QIDS points per week. This indicates a clinically meaningful improvement in depressive symptoms among the study participants. Notably, race significantly impacted QIDS scores, with Black individuals and those from other racial backgrounds exhibiting higher scores compared to the reference group, even after accounting for other variables. Additionally, a family history of psychiatric conditions was associated with higher QIDS scores.

However, this analysis is not without limitations. Firstly, while the model accounted for several demographic and clinical variables, it may not have captured all relevant factors influencing treatment response. Unmeasured confounders, such as socioeconomic status, specific medication adherence, or other concurrent treatments, could have influenced the results. Secondly, the study population may not be representative of all individuals with major depressive disorder, limiting the generalizability of the findings. Thirdly, the correlational nature of this analysis precludes causal inferences.

Future research directions should focus on addressing these limitations. Longitudinal studies with larger, more diverse samples are needed to confirm these findings and explore potential causal relationships. Incorporating measures of socioeconomic status, medication adherence, and other relevant clinical variables could provide a more comprehensive understanding of treatment response. Additionally, investigating the biological mechanisms underlying the observed racial disparities in treatment outcomes could lead to more targeted and effective interventions. Finally, studies exploring the impact of different treatment modalities or combinations on depressive symptoms in diverse populations are warranted.

Appendix A Dataset Description

Variable	File	Description
ID	All	Subject Identifier
SEX	STARD Demo 2	Gender (1 = Male, 2 = Female)
HSPNC	STARD Demo 2	Is the patient Hispanic (1 = yes, 0 = no)
WHITE	STARD Demo 2	Race (White); 1 = yes, 0 = no
BLACK	STARD Demo 2	Race (Black); 1 = yes, 0 = no
ASIAN	STARD Demo 2	Race (Asian); 1 = yes, 0 = no
AMIND	STARD Demo 2	Race (American Indian); 1 = yes, 0 = no
HAWAI	STARD Demo 2	Race (Hawaiian); 1 = yes, 0 = no
dage	STARD Demo 1	Age at onset of first major depressive episode
epino	STARD Demo 1	Number of major depressive episodes
episode_date	STARD Demo 1	Length of current major depressive episode
DEP	STARD Demo 1	Family history of depression (1 = yes, 0 = no, -3= unknown)
BIP	STARD Demo 1	Family history of bipolar disorder (1 = yes, 0 = no, -3= unknown)
ALCOHOL	STARD Demo 1	Family history of alcohol abuse (1 = yes, 0 = no, -3= unknown)
DRUG_PHX	STARD Demo 1	Family history of drug abuse (1 = yes, 0 = no, -3= unknown)
SUIC_PHX	STARD Demo 1	Family history of suicide (1 = yes, 0 = no, -3= unknown)
QSTOT	STARD QIDS	Total QIDS score
DATE	STARD QIDS	Number of days in study
WEEK	STARD QIDS	Number of weeks in study

Appendix B Models

Model 1 (Baseline Model):

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: QSTOT ~ 1 + WEEK + (1 + WEEK | ID)
   Data: full_data
Control: lmerControl(optimizer = "Nelder_Mead")

     AIC      BIC   logLik deviance df.resid
 93562.5  93608.8 -46775.3  93550.5    16398

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.5626 -0.5969 -0.0126  0.5490  3.8109

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 ID       (Intercept) 12.30965 3.5085
          WEEK         0.07854 0.2803   0.02
 Residual             10.85096 3.2941
Number of obs: 16404, groups:  ID, 3304

Fixed effects:
             Estimate Std. Error t value
(Intercept) 13.543634   0.073826  183.45
WEEK        -0.590902   0.008373  -70.57

Correlation of Fixed Effects:
     (Intr)
WEEK -0.334
```

Model 2 (Extended Model):

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: QSTOT ~ 1 + WEEK + SEX + Race_Collapsed + dage +
epino + Any_Fam_Psyc_Hist +
    (1 + WEEK | ID)
   Data: full_data
Control: lmerControl(optimizer = "Nelder_Mead")

     AIC      BIC   logLik deviance df.resid
 80513.3  80611.6 -40243.7  80487.3    14137

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2058 -0.5899 -0.0174  0.5490  3.8817

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 ID       (Intercept) 11.81240 3.4369
          WEEK         0.07926 0.2815   0.01
 Residual             10.74669 3.2782
Number of obs: 14150, groups:  ID, 2835

Fixed effects:
                     Estimate Std. Error t value
(Intercept)         10.647511   0.624952  17.037
WEEK                -0.596946   0.008995 -66.364
SEX                  0.534973   0.154045   3.473
Race_CollapsedBlack  3.373766   0.572378   5.894
Race_CollapsedOther  3.885554   0.817017   4.756
Race_CollapsedWhite  1.805103   0.547643   3.296
dage                -0.010921   0.005280  -2.069
epino                0.010172   0.008213   1.239
Any_Fam_Psyc_Hist    0.214410   0.166278   1.289

Correlation of Fixed Effects:
            (Intr) WEEK   SEX    Rc_ClB Rc_ClO Rc_ClW dage
epino
WEEK        -0.042
SEX         -0.425 -0.001
Rc_CllpsdBl -0.791  0.000 -0.008
Rc_CllpsdOt -0.551 -0.001 -0.019  0.627
Rc_CllpsdWh -0.838  0.000  0.012  0.935  0.655
dage        -0.261 -0.003  0.097 -0.065 -0.022 -0.044
epino       -0.143 -0.001  0.148 -0.060 -0.021 -0.033  0.212
Any_Fm_Ps_H -0.145  0.001 -0.064 -0.036 -0.046 -0.060  0.127
-0.015
```

Model 3 (Custom Model):

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula:
QSTOT ~ 1 + WEEK + Total_Fam_Psyc_His + dage_scaled + epino_scaled
    Race_Collapsed + episode_date_scaled + (1 + WEEK | ID)
   Data: full_data
Control: lmerControl(optimizer = "Nelder_Mead")

     AIC      BIC   logLik deviance df.resid
 80240.7  80338.9 -40107.4  80214.7    14089

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2041 -0.5897 -0.0169  0.5502  3.8716

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 ID       (Intercept) 11.80991 3.4366
          WEEK         0.07918 0.2814   0.00
 Residual             10.75821 3.2800
Number of obs: 14102, groups:  ID, 2824

Fixed effects:
                      Estimate Std. Error t value
(Intercept)          11.328305   0.542494  20.882
WEEK                 -0.597739   0.009012 -66.326
Total_Fam_Psyc_His    0.219989   0.068535   3.210
dage_scaled          -0.132584   0.076402  -1.735
epino_scaled          0.079358   0.075785   1.047
Race_CollapsedBlack   3.204438   0.573384   5.589
Race_CollapsedOther   3.657898   0.818422   4.469
Race_CollapsedWhite   1.639155   0.548049   2.991
episode_date_scaled   0.207475   0.073082   2.839

Correlation of Fixed Effects:
            (Intr) WEEK   T_F_P_ dg_scl epn_sc Rc_ClB Rc_ClO Rc_ClW
WEEK        -0.051
Ttl_Fm_Ps_H -0.084  0.002
dage_scaled  0.018 -0.003  0.173
epino_scald  0.042 -0.001 -0.020  0.204
Rc_CllpsdBl -0.934  0.000 -0.061 -0.073 -0.062
Rc_CllpsdOt -0.651 -0.001 -0.079 -0.031 -0.023  0.628
Rc_CllpsdWh -0.974  0.000 -0.083 -0.054 -0.037  0.935  0.657
epsd_dt_scl  0.035 -0.002 -0.019  0.078  0.103 -0.050 -0.050 -0.030
```

Fitted Values:

```
  [1] 10.02342311 11.65668004  8.39016618 13.28993697  2.70929346  7.19064033 10.17820492  8.68442262 11.67198721  4.94996690 10.57921604  8.10446887 11.81658962 10.51734736  8.18891768 12.57579577  6.39709424
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 [35] 12.20540217 10.29963778  8.04244822 15.56641342 14.06162038 12.55682734 17.07120646 13.43544491  8.89083213  6.61852574 10.40570306  4.34621936  0.82610614  9.32373503  6.49119207 10.74000651  4.36678484
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[103] 15.06717573 16.73943398 15.90330485 17.57556310  5.01677751  3.23130002  7.99257334 10.37321000  6.80225501 17.81283627 15.69051722 12.00041546  6.17969900 10.54523634 14.91077369  8.36246767 13.45559457
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[137]  7.17380045 11.37450054  8.97410049 10.84198341 13.46053949 12.41311706 15.55538436  7.03253737 11.21855797  4.93952707  1.45117657  9.82321777  8.55013361 12.28141283  6.01510111  9.00797726  4.51866303
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[171]  7.64135452  3.36236978  6.21502627  1.22287740 17.91048693 16.80547176 17.87365309 20.12051726 19.38384048 18.77462829 16.65695156 15.37465236 11.12666822 12.54266293 11.63514927  8.79798476  9.21152097
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[800] 13.32290522 16.57330623  9.02627654 11.64616442  7.71633260  5.75141668 10.33622048 19.05397438 17.17722898 18.13848882 17.22300326 16.30751770 14.93428936 11.96420913 10.61572338 13.31269487 12.49331695
[817] 12.55894906  9.93366458  1.59886381  4.26326141  6.03952648 -1.06553380  5.95071323  9.59205662 12.45248245  9.55768174 11.48754888  6.66288103  8.11028138 10.52261531 20.26846960 18.37965092 17.43524158
[834] 19.32406026  8.32013418  7.21830709  1.70917160  6.11647999  5.01465289  3.36191225 10.33228020  9.38811931 11.27644110  7.97187797 12.22060199  4.26042403 10.54205360  8.23613895 10.62156790  5.85071000
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```

Model Metrics:

Modnames <chr>	K <dbl>	AICc <dbl>	Delta_AICc <dbl>	AICcWt <dbl>	Eratio <dbl>
M1	6	93562.55	13321.81	0	Inf
M2	13	80513.34	272.60	0	1.560832e+59
M3	13	80240.74	0.00	1	1.000000e+00

Appendix C Code

```
---
title: "DA2"
format:
  html:
    toc: true
    embed-resources: true
    code-fold: true
    code-line-numbers: true
    code-summary: "SHOW ME THE CODE!"
    code-tools: true
editor: visual
---


```{r setup}
#| message: false
#| warning: false
library(lme4)
library(tidyverse)
library(AICcmodavg)
library(gtsummary)
library(dplyr)
library(e1071)
```


```{r}
# read data
```
```

```
demo1 <- read.csv("C:\\Users\\jasmi\\OneDrive - Southern Methodist University\\6302\\Data Assignments\\DA2\\STARD Demo 1.csv")

demo2 <- read.csv("C:\\Users\\jasmi\\OneDrive - Southern Methodist University\\6302\\Data Assignments\\DA2\\STARD Demo 2.csv")

starQids <- read.csv("C:\\Users\\jasmi\\OneDrive - Southern Methodist University\\6302\\Data Assignments\\DA2\\STARD QIDS.csv")
```

Part I

a)

```{r}
# merge demograph datasets

demo <- merge(demo1, demo2, by = "ID")
```

```{r}
# clean data

colMeans(is.na(demo))

d2 <- demo[complete.cases(demo),]

colMeans(is.na(d2))

colMeans(is.na(starQids))

sq <- starQids[complete.cases(starQids),]

colMeans(is.na(sq))
```

b)

```{r}
# remove anything with .x

qids.3 <- starQids[-grep("\\.", starQids$WEEK), ]
```

```
# count of subjects in qids.3
length(unique(qids.3$ID))

# Get maximum week per subject
max_weeks <- aggregate(WEEK ~ ID, data = qids.3, max)

# Convert max week to number of visits
max_weeks$Visits <- (max_weeks$WEEK / 2) + 1

# Filter subjects with at least 3 visits
qids.3 <- qids.3[qids.3$ID %in% max_weeks$ID[max_weeks$Visits >= 3], ]

# Create Table 1: Count of subjects by number of visits
table_1 <- table(max_weeks$Visits[max_weeks$Visits >= 3])
print(table_1)

# count of subjects in qids.3 after filtering subjects with >=3 visits
length(unique(qids.3$ID))
```

```{r}
library(gt)

# Convert table to a data frame
df_table <- as.data.frame(table(max_weeks$Visits[max_weeks$Visits >= 3]))
colnames(df_table) <- c("Visits", "Frequency")

# Transpose the table
df_table_t <- as.data.frame(t(df_table))
colnames(df_table_t) <- df_table_t[1, ]  # Set column names as visits values
df_table_t <- df_table_t[-1, , drop = FALSE]  # Remove the first row (it was column names)

# Add "Visits" as a row label and "Frequency" as another row label
df_table_t <- rbind("Visits" = colnames(df_table_t), "Frequency" = df_table_t)
```

```
df_table_t <- as.data.frame(df_table_t)  # Convert back to a data frame

# Convert to gt table
df_table_t %>%
  gt(rownames_to_stub = TRUE) %>%
  tab_stubhead(label = " ")  # Adds a space instead of a stub column name
```

c)

```{r}
# Create a vector of unique IDs from QIDS.3
valid_ids <- unique(qids.3$ID)

# Filter the demo dataset to include only these subjects
demo_filtered <- demo %>% filter(ID %in% valid_ids)

# Check sample size
length(unique(demo_filtered$ID))

# Sample size is one less than qids.3

# Check for IDs in QIDS.3 that are missing in demo_filtered
#missing_in_demo <- setdiff(unique(qids.3$ID), demo_filtered$ID)
#missing_in_demo

#Turns out ID 2172 is missing in the demo dataset so we remove it from our dataset

# Replace invalid values (-5 for categorical, <0 for numeric)
demo_filtered <- demo_filtered %>% filter(ID != 2172)
```

```
# Replace both -3 and -5 with NA for variables where these values represent missing/invalid data
demo_filtered$DEP[demo_filtered$DEP == -3 | demo_filtered$DEP == -5] <- NA
demo_filtered$BIP[demo_filtered$BIP == -3 | demo_filtered$BIP == -5] <- NA
demo_filtered$ALCOHOL[demo_filtered$ALCOHOL == -3 | demo_filtered$ALCOHOL == -5] <- NA
demo_filtered$DRUG_PHX[demo_filtered$DRUG_PHX == -3 | demo_filtered$DRUG_PHX == -5] <- NA
demo_filtered$SUIC_PHX[demo_filtered$SUIC_PHX == -3 | demo_filtered$SUIC_PHX == -5] <- NA

# Replace all negative values with NA for epino, episode_date and dage
demo_filtered$HSPNC[demo_filtered$HSPNC < 0] <- NA
demo_filtered$epino[demo_filtered$epino < 0] <- NA
demo_filtered$dage[demo_filtered$dage < 0] <- NA


# Since all episode_date values are negative, we will make them positive
demo_filtered$episode_date <- abs(demo_filtered$episode_date)
```

```{r}
# Histogram of 'dage'
ggplot(demo_filtered, aes(x = dage)) +
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7) +
  labs(title = "Histogram of Age (dage)", x = "Age", y = "Frequency")

# Histogram of 'epino'
ggplot(demo_filtered, aes(x = epino)) +
  geom_histogram(bins = 30, fill = "green", alpha = 0.7) +
  labs(title = "Histogram of Episodes (epino)", x = "Number of Episodes", y = "Frequency")

# Histogram of 'episode_date'
ggplot(demo_filtered, aes(x = episode_date)) +
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7) +
  labs(title = "Distribution of Episode Date", x = "Episode Date", y = "Frequency")
```

```
# Calculate skewness while ignoring NAs
skewness(demo_filtered$dage, na.rm = TRUE)
skewness(demo_filtered$epino, na.rm = TRUE)
skewness(demo_filtered$episode_date, na.rm = TRUE)
```

```{r}
# Classify race categories # THIS NEEDS TO BE SPECIFIED IN THE REPORT
demo_filtered <- demo_filtered %>%
  mutate(
    # White race classification
    Race_White = ifelse(WHITE == 1, "White", NA),

    # Black race classification
    Race_Black = ifelse(BLACK == 1 & !(WHITE == 1), "Black", NA),

    # Asian race classification, only for those who did not select Black
    Race_Asian = ifelse(ASIAN == 1 & !(WHITE == 1 | BLACK == 1), "Asian", NA),

    # Other race classification: for subjects who did not select White, Black, or Asian
    Race_Other = ifelse(
      (HSPNC == 1 | AMIND == 1 | HAWAI == 1) & !(WHITE == 1 | BLACK == 1 | ASIAN == 1),
      "Other",
      NA
    )
  )

# Summary for continuous variables (Median and IQR for skewed data)
summary_continuous <- demo_filtered %>%
  summarise(
```

```
  dage_median = median(dage, na.rm = TRUE),
  dage_IQR = IQR(dage, na.rm = TRUE),
  episode_date_median = median(episode_date, na.rm = TRUE),
  episode_date_IQR = IQR(episode_date, na.rm = TRUE),
  epino_median = median(epino, na.rm = TRUE),
  epino_IQR = IQR(epino, na.rm = TRUE)
 )

# Summary for categorical variables (N and percentages for each category)
summary_categorical <- demo_filtered %>%
 summarise(
  SEX_Female_n = sum(SEX == 2, na.rm = TRUE),  # Count for Female
  SEX_Female_pct = sum(SEX == 2, na.rm = TRUE) / n() * 100,  # Percentage for Female
  SEX_Male_n = sum(SEX == 1, na.rm = TRUE),  # Count for Male
  SEX_Male_pct = sum(SEX == 1, na.rm = TRUE) / n() * 100,  # Percentage for Male

  DEP_No_n = sum(DEP == 0, na.rm = TRUE),  # Count for No depression history
  DEP_No_pct = sum(DEP == 0, na.rm = TRUE) / n() * 100,  # Percentage for No depression history
  DEP_Yes_n = sum(DEP == 1, na.rm = TRUE),  # Count for Yes depression history
  DEP_Yes_pct = sum(DEP == 1, na.rm = TRUE) / n() * 100,  # Percentage for Yes depression history

  ALCOHOL_No_n = sum(ALCOHOL == 0, na.rm = TRUE),  # Count for No alcohol use
  ALCOHOL_No_pct = sum(ALCOHOL == 0, na.rm = TRUE) / n() * 100,  # Percentage for No alcohol use
  ALCOHOL_Yes_n = sum(ALCOHOL == 1, na.rm = TRUE),  # Count for Yes alcohol use
  ALCOHOL_Yes_pct = sum(ALCOHOL == 1, na.rm = TRUE) / n() * 100,  # Percentage for Yes alcohol use

  DRUG_PHX_No_n = sum(DRUG_PHX == 0, na.rm = TRUE),  # Count for No drug use
  DRUG_PHX_No_pct = sum(DRUG_PHX == 0, na.rm = TRUE) / n() * 100,  # Percentage for No drug use
  DRUG_PHX_Yes_n = sum(DRUG_PHX == 1, na.rm = TRUE),  # Count for Yes drug use
  DRUG_PHX_Yes_pct = sum(DRUG_PHX == 1, na.rm = TRUE) / n() * 100,  # Percentage for Yes drug use

  SUIC_PHX_No_n = sum(SUIC_PHX == 0, na.rm = TRUE),  # Count for No suicidal history
  SUIC_PHX_No_pct = sum(SUIC_PHX == 0, na.rm = TRUE) / n() * 100,  # Percentage for No suicidal history
```

```
  SUIC_PHX_Yes_n = sum(SUIC_PHX == 1, na.rm = TRUE),  # Count for Yes suicidal history
  SUIC_PHX_Yes_pct = sum(SUIC_PHX == 1, na.rm = TRUE) / n() * 100,  # Percentage for Yes suicidal history

  # Race Summary
  Race_White_n = sum(Race_White == "White", na.rm = TRUE),  # Count for White
  Race_White_pct = sum(Race_White == "White", na.rm = TRUE) / n() * 100,  # Percentage for White

  Race_Black_n = sum(Race_Black == "Black", na.rm = TRUE),  # Count for Black
  Race_Black_pct = sum(Race_Black == "Black", na.rm = TRUE) / n() * 100,  # Percentage for Black

  Race_Asian_n = sum(Race_Asian == "Asian", na.rm = TRUE),  # Count for Asian
  Race_Asian_pct = sum(Race_Asian == "Asian", na.rm = TRUE) / n() * 100,  # Percentage for Asian

  Race_Other_n = sum(Race_Other == "Other", na.rm = TRUE),  # Count for Other
  Race_Other_pct = sum(Race_Other == "Other", na.rm = TRUE) / n() * 100  # Percentage for Other
 )

# 4. Combine continuous and categorical summaries into a final table
summary_table <- bind_cols(summary_continuous, summary_categorical)

# 5. Print the summary table
print(summary_table)
```

This table can be used to make a nicer table in word/pdf for the report. Look at the STARD paper for inspo. page 4 of 13.

```
```{r}
library(gt)
library(dplyr)
library(tidyr)
library(stringr)
```
```

```
# Assuming summary_table is your data frame

# Create a clean mapping of variable names
name_mapping <- c(
  "SEX_Female_n" = "Female_N",
  "SEX_Female_pct" = "Female_Percentage",
  "SEX_Male_n" = "Male_N",
  "SEX_Male_pct" = "Male_Percentage",

  "DEP_No_n" = "No_N",
  "DEP_No_pct" = "No_Percentage",
  "DEP_Yes_n" = "Yes_N",
  "DEP_Yes_pct" = "Yes_Percentage",

  "ALCOHOL_No_n" = "No_N",
  "ALCOHOL_No_pct" = "No_Percentage",
  "ALCOHOL_Yes_n" = "Yes_N",
  "ALCOHOL_Yes_pct" = "Yes_Percentage",

  "DRUG_PHX_No_n" = "No_N",
  "DRUG_PHX_No_pct" = "No_Percentage",
  "DRUG_PHX_Yes_n" = "Yes_N",
  "DRUG_PHX_Yes_pct" = "Yes_Percentage",

  "SUIC_PHX_No_n" = "No_N",
  "SUIC_PHX_No_pct" = "No_Percentage",
  "SUIC_PHX_Yes_n" = "Yes_N",
  "SUIC_PHX_Yes_pct" = "Yes_Percentage",

  "Race_White_n" = "White_N",
  "Race_White_pct" = "White_Percentage",
  "Race_Black_n" = "Black_N",
  "Race_Black_pct" = "Black_Percentage",
```

```
  "Race_Asian_n" = "Asian_N",
  "Race_Asian_pct" = "Asian_Percentage",
  "Race_Other_n" = "Other_N",
  "Race_Other_pct" = "Other_Percentage"
)

# Convert summary_table into a long format with separate N and %
long_table <- summary_table %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value") %>%
  mutate(
    Category = case_when(
      str_detect(Variable, "Race_") ~ "Race",
      str_detect(Variable, "SEX_") ~ "Gender",
      str_detect(Variable, "DEP_") ~ "Depression History",
      str_detect(Variable, "ALCOHOL_") ~ "Alcohol Use",
      str_detect(Variable, "DRUG_PHX_") ~ "Drug Use",
      str_detect(Variable, "SUIC_PHX_") ~ "Suicidal History",
      TRUE ~ "Other"
    ),
    # Rename variables and create N and Percentage identifiers
    Variable = recode(Variable, !!!name_mapping),
    Value_Type = ifelse(str_detect(Variable, "_N"), "N", "Percentage"),
    Variable = str_replace(Variable, "_(N|Percentage)", "") #Remove _N and _Percentage
  ) %>%
  pivot_wider(names_from = Value_Type, values_from = "Value") %>%
  filter(Category != "Other") # Remove the "Other" category

# Define category order
category_order <- c("Race", "Gender", "Depression History", "Alcohol Use", "Drug Use", "Suicidal History")

# Format the table using `gt()`
long_table %>%
  mutate(Category = factor(Category, levels = category_order)) %>% #order categories
```

```
  arrange(Category, Variable) %>%
  gt(groupname_col = "Category") %>%
  tab_header(title = md("**Baseline Characteristics of Patients**")) %>%
  cols_label(
    Variable = "Characteristic",
    N = "N",
    Percentage = "%"
  ) %>%
  fmt_number(columns = c(N, Percentage), decimals = 1) %>%
  tab_options(
    table.font.size = "small",
    heading.title.font.size = "medium",
    column_labels.font.weight = "bold"
  )
```

```{r}
library(gt)
library(dplyr)

# Assuming you have a data frame called 'other_stats' with IQR and Median values
# Example other_stats (replace with your actual data)
other_stats <- data.frame(
  Characteristic = c("Age", "Major Depressive Episodes", "Length of Current MDE"),
  IQR = c(21.0, 4.0, 641.0),
  Median = c(21.0, 3.0, 247.0)
)

# Create the table using gt()
other_stats %>%
  gt() %>%
  cols_label(
```

```
  Characteristic = "Characteristic",
  IQR = "IQR",
  Median = "Median"
 ) %>%
 fmt_number(columns = c(IQR, Median), decimals = 1) %>%
 tab_options(
  table.font.size = "small",
  heading.title.font.size = "medium",
  column_labels.font.weight = "bold"
 )
```

d)

In this analysis, subjects were allowed to select multiple races, and it was necessary to classify them into broader categories for reporting purposes. The following classification rules were applied to categorize race:

1. **White**: Subjects who identified as **White** were classified as **White**.
2. **Black**: Subjects who identified as **Black** and did **not** identify as **White** were classified as **Black**. This ensures that individuals who selected both **White** and **Black** are not mistakenly categorized as **Black**.
3. **Asian**: Subjects who identified as **Asian** and did **not** identify as either **Black** or **White** were classified as **Asian**.
4. **Other**: Subjects who selected races such as **Hispanic**, **AMIND**, or **HAWAI**, and did **not** select **White**, **Black**, or **Asian**, were classified as **Other**.

This classification approach ensures that individuals who selected multiple races are categorized based on their most specific race (e.g., **Black** if **Black** and **White** were selected). Additionally, the **Other** category includes subjects who selected races outside the main classifications, ensuring that their race is still represented in the dataset without misclassification.

e)

```{r}
# Create the new composite variable 'Any.Fam.Psyc.Hist'
demo_filtered <- demo_filtered %>%
```

```
  mutate(
    Any_Fam_Psyc_Hist = ifelse(DEP == 1 | BIP == 1 | ALCOHOL == 1 | DRUG_PHX == 1 | SUIC_PHX == 1, 1, 0)
  )
```

Part II

a)

```{r}
library(ggplot2)
library(dplyr)

set.seed(111)

# Randomly select 100 unique subjects
rand_ids <- sample(unique(starQids$ID), size = 100, replace = FALSE)

# Filter dataset to only include the selected subjects
rand_samp <- starQids %>% filter(ID %in% rand_ids)

# Create the spaghetti plot
g1 <- ggplot(data = rand_samp, aes(x = WEEK, y = QSTOT, group = ID, color = factor(ID))) +
  geom_line(alpha = 0.5) +  # Reduce opacity for clarity
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week") +
  labs(title = "Spaghetti Plot of QIDS Scores Over Time", color = "Subject ID") +
  theme_bw() +  # Use a clean theme
  theme(legend.position = "none")  # Hide legend for readability

# Print the plot
print(g1)
```

```

## b)

```{r}

# Jitter week variable
starQids$jweek <- jitter(starQids$WEEK)

# Create the loess curve plot
g_loess <- ggplot(data = starQids, aes(x = jweek, y = QSTOT)) +
  geom_jitter(alpha = 0.3, color = "blue") +  # Jittered points for visibility
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week", breaks = 0:16) +
  stat_smooth(method = "loess", se = FALSE, color = "red", size = 1.2) +  # Loess smoothing
  labs(title = "Loess Curve of QIDS Scores Across All Subjects") +
  theme_minimal()

# Print the plot
print(g_loess)

```

c)

```{r}
# Merge gender info with QIDS dataset
sub_cols <- demo_filtered[, c("SEX", "ID")]
sq_gen <- merge(x = starQids, y = sub_cols, by = "ID")

# Check for missing values
```
```

```
colSums(is.na(sq_gen))

# Jitter week values for better visualization
sq_gen$jweek <- jitter(sq_gen$WEEK)

# Create faceted plot by gender
g_gender <- ggplot(data = sq_gen, aes(x = jweek, y = QSTOT, color = factor(SEX))) +
  geom_jitter(alpha = 0.3) +  # Add jitter for better visibility
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week", breaks = 0:16) +
  stat_smooth(method = "loess", se = FALSE, size = 1.2) +  # Loess curve for each facet
  facet_wrap(~SEX, labeller = labeller(SEX = c("1" = "Male", "2" = "Female"))) +
  labs(title = "QIDS Scores Over Time by Gender", color = "Gender") +
  theme_minimal()

# Print the plot
print(g_gender)

```
```
## d)

```{r}
# Create a single Race_Collapsed variable
demo_filtered <- demo_filtered %>%
  mutate(
    Race_Collapsed = case_when(
      Race_White == "White" ~ "White",
      Race_Black == "Black" ~ "Black",
      Race_Asian == "Asian" ~ "Asian",
      Race_Other == "Other" ~ "Other",
      TRUE ~ NA_character_  # Assign NA for any missing values
    )
```
```

```
 )

# Merge race info with QIDS dataset
sub_eth <- demo_filtered[, c("ID", "Race_Collapsed")]
sq_eth <- merge(starQids, sub_eth, by = "ID")

# Jitter week values for better visualization
sq_eth$jweek <- jitter(sq_eth$WEEK)

# Create faceted plot by ethnicity
g_ethn <- ggplot(data = sq_eth, aes(x = jweek, y = QSTOT, color = factor(Race_Collapsed))) +
  geom_jitter(alpha = 0.3) +  # Reduce overplotting
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week", breaks = 0:16) +
  stat_smooth(method = "loess", se = FALSE, size = 1.2) +  # Add loess curve for each category
  facet_wrap(~Race_Collapsed) +
  labs(title = "QIDS Scores Over Time by Ethnicity", color = "Ethnicity") +
  theme_minimal()

print(g_ethn)
```

e)

```{r}
# Merge 'epino' data with QIDS dataset
sub_episodes <- demo_filtered[, c("ID", "epino")]
sq_epi <- merge(starQids, sub_episodes, by = "ID")

# Ensure 'epino' has valid values before binning
sq_epi <- sq_epi %>% filter(!is.na(epino))  # Remove NAs if needed
```

```
# Bin 'epino' into 3 groups using cut_number
sq_epi$epi2 <- cut_number(sq_epi$epino, n = 3, labels = c("Low", "Medium", "High"))

# Create faceted plot by episode binning
g_episode <- ggplot(data = sq_epi, aes(x = WEEK, y = QSTOT)) +
  geom_jitter(alpha = 0.3, color = "blue") +  # Add jitter for better visualization
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week", breaks = 0:16) +
  stat_smooth(method = "loess", se = FALSE, color = "red", size = 1.2) +  # Loess curve
  facet_wrap(~epi2) +  # Facet by episode binning
  labs(title = "QIDS Scores Over Time by Episode Count", x = "Week", y = "QIDS Score") +
  theme_minimal()

print(g_episode)

```

f)

```{r}

# Merge gender (`SEX`) with episode-binned data (`sq_epi`)
sq_epi_sex <- merge(sq_epi, sub_cols, by = "ID")

# Check for missing values
colSums(is.na(sq_epi_sex))

# Create faceted plot by gender with separate loess curves for each episode bin
g_gender_epi <- ggplot(data = sq_epi_sex, aes(x = WEEK, y = QSTOT, color = epi2)) +
  geom_jitter(alpha = 0.3) +  # Reduce overplotting
  scale_y_continuous(name = "QIDS Score") +
  scale_x_continuous(name = "Week", breaks = 0:16) +
```

```
 stat_smooth(method = "loess", se = FALSE, size = 1) +  # Loess curve for each bin
 facet_wrap(. ~ SEX, labeller = labeller(SEX = c("1" = "Male", "2" = "Female"))) +  # Gender labels
 labs(
   title = "QIDS Scores Over Time by Gender & Episode Count",
   color = "Episode Bins"
 ) +
 theme_minimal()

# Print the plot
print(g_gender_epi)
```

Part III

a)

```{r}
full_data <- merge(sq, demo_filtered, by="ID")
lmer.1 <- lmer(QSTOT ~ 1 + WEEK + (1 + WEEK | ID),
           data = full_data,
           REML = F,
           control = lmerControl(optimizer = "Nelder_Mead"))
summary(lmer.1)
```

You can use this for the report:

Interpreting Your Model Output (`lmer.1`)

Key results: - **Fixed Effects:** - **Intercept (13.54)** → The estimated **QIDS score at Week 0** is **13.54**. - **WEEK (-0.59)** → On average, **QIDS scores decrease by 0.59 per week**. - **t-value for WEEK (-70.57)** is large, meaning **WEEK has a strong effect**.

- **Random Effects:**
  - **Intercept variance (12.31, SD = 3.51)** → Some subjects start with much higher or lower QIDS scores.
  - **Slope variance for WEEK (0.078, SD = 0.28)** → Subjects vary slightly in their rate of improvement.
  - **Residual variance (10.85, SD = 3.29)** → This represents within-subject variation in QIDS scores.
  - **Intercept-Slope Correlation (0.02)** → Essentially **no correlation** between baseline QIDS score and rate of improvement.
- **Model Fit:**
  - **AIC = 93562.5, BIC = 93608.8** → These will be used to compare with the expanded model.

b)

```{r}
full_data$Race_Collapsed <- as.factor(full_data$Race_Collapsed)

lmer.2 <- lmer(QSTOT ~ 1 + WEEK + SEX + Race_Collapsed + dage + epino + Any_Fam_Psyc_Hist + (1 + WEEK | ID),
        data = full_data,
        REML = F,
        control = lmerControl(optimizer = "Nelder_Mead"))
summary(lmer.2)
```

You can use this for the report:

Model Summary: - **Fixed Effects:** - **Intercept (10.65)**: This is the estimated **QIDS score** at **Week 0** (baseline).\

- **WEEK (-0.60)**: On average, the **QIDS score decreases by 0.60 per week**, suggesting that depression severity decreases over time. - **SEX (0.53)**: Males have **0.53 higher QIDS scores** than females, on average. - **Race_Collapsed**:\

- **Black (3.37)**: Black subjects tend to have **3.37 higher QIDS scores** than the reference category (likely "White").\

- **Other (3.89)**: Subjects from other racial backgrounds tend to have **3.89 higher QIDS scores** than the reference category.\

- **White (1.81)**: White subjects tend to have **1.81 higher QIDS scores** compared to the reference group. - **dage (-0.01)**: For every one-year increase in **age**, the **QIDS score decreases slightly by 0.01**. - **epino (0.01)**: **Epilepsy history** has a **small positive effect** on QIDS scores, but it's not significant at the conventional threshold (t value = 1.24).\

- **Any_Fam_Psyc_Hist (0.21)**: Family psychiatric history is positively associated with **QIDS score**, but not statistically significant at the conventional threshold (t value = 1.29).

- **Random Effects**:
  - **Intercept Variance (11.81, SD = 3.44)**: There is **substantial variability** in baseline QIDS scores across subjects.\
  - **Slope Variance for WEEK (0.08, SD = 0.28)**: The rate of change in QIDS scores varies moderately between subjects.
  - **Residual Variance (10.85, SD = 3.29)**: There is still considerable unexplained variability within subjects.
- **Correlation of Fixed Effects**:
  - **Correlation between SEX and Race_Collapsed** is somewhat negative, which may suggest that gender and race have some correlation, but it's **not very strong**.
  - The **correlation between `dage` and other predictors** is relatively small, indicating little to no collinearity issues.

Key Takeaways:

1. **WEEK has a strong negative effect** on QIDS scores, meaning participants' depression symptoms decrease over time.
2. **Race** appears to have a large effect on QIDS scores, with subjects in the **Other** and **Black** categories showing **higher QIDS scores** on average compared to the reference group.
3. **No meaningful correlation** between baseline QIDS scores and rate of change over time (random intercept and slope correlation is near 0).
4. **Family psychiatric history (Any_Fam_Psyc_Hist)** does not show a statistically significant effect on QIDS scores.

Comparison of Model Fit (lmer.1 vs. lmer.2)

Model 1 (lmer.1):

- **AIC**: 93,562.5\
- **BIC**: 93,608.8\
- **logLik**: -46,775.3\
- **Deviance**: 93,550.5\
- **Residual Variance (Std. Dev.)**: 10.85 (3.29)

Model 2 (lmer.2):

- **AIC**: 80,513.3\
- **BIC**: 80,611.6\
- **logLik**: -40,243.7\
- **Deviance**: 80,487.3\
- **Residual Variance (Std. Dev.)**: 10.75 (3.28)

Interpretation:

1. **AIC** and **BIC**:
   - **lmer.2** has a significantly **lower AIC (80,513.3)** and **BIC (80,611.6)** compared to **lmer.1** (AIC = 93,562.5 and BIC = 93,608.8).\
   - A lower AIC and BIC indicate **better model fit** in **lmer.2** after including the additional predictors.
2. **logLik**:
   - The **log-likelihood** of **lmer.2 (-40,243.7)** is also higher than **lmer.1 (-46,775.3)**, suggesting that the additional predictors in **lmer.2** improve the fit by explaining more variation in the data.
3. **Residual Variance**:
   - The residual variance in **lmer.2** (10.75) is slightly lower than **lmer.1** (10.85), suggesting that **lmer.2** accounts for more of the unexplained variance.

Conclusion:

- The **inclusion of additional predictors (SEX, Race_Collapsed, dage, epino, Any.Fam.Psyc.Hist)** in **lmer.2** improves the model fit, as evidenced by the **lower AIC, BIC**, and **higher log-likelihood**.
- This suggests that **these predictors** are likely **meaningful** in explaining variation in **QIDS scores** over time.

c)

```{r}

full_data <- full_data %>%
  mutate(
    dage_scaled = scale(dage),
    epino_scaled = scale(epino),
```

```
  episode_date_scaled = scale(episode_date),
  Total_Fam_Psyc_His = DEP + DRUG_PHX + SUIC_PHX + BIP + ALCOHOL
 )

lmer.3 <- lmer(QSTOT ~ 1 + WEEK + Total_Fam_Psyc_His +
             dage_scaled + epino_scaled + Race_Collapsed + episode_date_scaled +
             (1 + WEEK | ID),
           data = full_data,
           REML = F,
           control = lmerControl(optimizer = "Nelder_Mead"))
summary(lmer.3)
```

```
```

```{r}
# Get the fixed effects coefficients
fixed_effects <- coef(summary(lmer.3))

# Print the fixed effects coefficients
print(fixed_effects)

#if you only want the estimates.
estimates <- fixed_effects[,"Estimate"]
print(estimates)

#if you want the confidence intervals
confint(lmer.3)

#if you want the random effects.
ranef(lmer.3)

# If you want to put all of the coefficient data into a data frame.
fixed_effects_df <- as.data.frame(fixed_effects)
print(fixed_effects_df)
```

```

# Part IV

## a)

```{r}
mynames <- paste("M", as.character(1:3), sep = "")
mymodels <- list(lmer.1, lmer.2, lmer.3)
myaicc <- as.data.frame(aictab(cand.set = mymodels,
                               modnames = mynames,
                               sort = FALSE)[,-c(5,7)])

myaicc$Eratio <- max(myaicc$AICcWt) / myaicc$AICcWt

data.frame(Modnames = myaicc$Modnames, round(myaicc[,-1], 2))

```

Comparison of Model Fit:

Model 1: lmer.2 - **AIC**: 80,513.3\
- **BIC**: 80,611.6\
- **logLik**: -40,243.7\
- **Deviance**: 80,487.3\
- **Residual Variance (SD)**: 10.75 (3.28)

Model 2: lmer.3 - **AIC**: 80,240.7\
- **BIC**: 80,338.9\
- **logLik**: -40,107.4\
- **Deviance**: 80,214.7\
- **Residual Variance (SD)**: 10.76 (3.28)
```

Key Insights:

1. **AIC and BIC**:

   - **`lmer.3` has a lower AIC (80,240.7) and BIC (80,338.9)** compared to **`lmer.2` (AIC = 80,513.3, BIC = 80,611.6)**. This suggests that **`lmer.3`** provides a **better fit** to the data, as both AIC and BIC are lower.

2. **logLik**:

   - **`lmer.3` has a higher log-likelihood (-40,107.4)** compared to **`lmer.2` (-40,243.7)**. This indicates that **`lmer.3`** explains more variance in the data.

3. **Residual Variance**:

   - **Residual variance is almost identical** between the two models (10.75 vs. 10.76), suggesting that **`lmer.3`** does not introduce significant extra variability compared to **`lmer.2`**.

Conclusion: - **`lmer.3` is the more plausible model** based on the lower AIC, BIC, and higher log-likelihood, suggesting that the inclusion of **`Total_Fam_Psyc_His`** as a composite variable improves the model fit without significantly increasing the residual variance.

1. **Extract Fitted Values for `lmer.3`:**

```{r}

# I was unable to get the fitted values due to missing values so I had to do the following:

full_data_complete <- full_data[complete.cases(full_data[, c("SEX", "Race_Collapsed", "dage_scaled", "epino_scaled",
                                                "Total_Fam_Psyc_His", "episode_date_scaled")]), ]

lmer.3 <- lmer(QSTOT ~ 1 + WEEK + Total_Fam_Psyc_His +
                dage_scaled + epino_scaled + Race_Collapsed + episode_date_scaled +
                (1 + WEEK | ID),
               data = full_data_complete,
               REML = F,
               control = lmerControl(optimizer = "Nelder_Mead"))
```

```
full_data_complete$fitted_values <- fitted(lmer.3)


full_data_complete$fitted_values
```

2. **Select a Random Sample of 25 Subjects:**

```{r}
set.seed(111)
random_sample_ids <- sample(unique(full_data_complete$ID), 25)
subset_data <- full_data_complete %>%
  filter(ID %in% random_sample_ids)
```

3. **Create the Plot:**

```{r}
ggplot(subset_data, aes(x = WEEK, y = QSTOT, group = ID)) +
  geom_point(color = "blue", alpha = 0.5) +  # Raw data points
  geom_line(aes(y = fitted_values), color = "red", size = 1) +  # Predicted values
  facet_wrap(~ID, scales = "free_y") +  # One panel per subject
  labs(title = "Observed vs. Predicted QIDS Scores by Subject",
       x = "Week",
       y = "QIDS Score") +
  theme_minimal()
```